Ian Leung

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Education

University of Waterloo — **Sep 2025 – Present**

Bachelor of Computer Science and Finance - Double Major

- President's Scholarship of Distinction (**$5000**), Undergraduate Achievement Award (**$1500**)
- Statistics Minor

Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Proofs, Financial Markets and Data Analytics

Technical Skills

Languages	Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash
Technologies	Temporal, PyTorch, Supabase, Firebase, MongoDB, React, Maven, Qt, OpenRouter, OpenCV
Developer Tools	Linux, Git, Docker, Jupyter, DigitalOcean, Modal, Render, Vercel, Unity, Android Studio

Work Experience

Roblox — **San Mateo, CA**

Software Engineer Intern | Go, Python — *Jun 2026 – Present*

- Building an automated CI/CD pipeline that deploys configuration changes across **5,000+** global network devices (servers) using **GitHub Actions** and **Temporal**, replacing a slow, manual, error-prone process
- Designing a **canary-first** rollout with automated health checks (**Pingmesh, VictoriaMetrics**) and staged, gated expansion, validating changes on a small device set before fleet-wide deployment to Roblox's **100M+** daily active users
- Implementing deployment safety controls including device **locking**, automatic **rollback**, and error classification, so devices self-recover to a known-good state on failure

Sunnybrook Research Institute — **Toronto, ON**

Software Research Intern | Python — *Jul 2025 – Aug 2025*

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework with **PyQt and Pydantic**, generating synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Implemented a logging API using **Difflib and Loguru** to validate test results against baselines, identifying **5+** bugs

Projects

Data Science Club Website | TypeScript, React, Supabase, HTML, CSS

- Maintained www.uwdatascience.ca, UWaterloo's data science club website serving **600+ members**
- Replaced the executive team's manual term-start **Google Form** with an in-app onboarding flow backed by a normalized **Supabase/Postgres** schema that autofills profile and role data from existing records

Pharmagotchi: Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a native **Android** (Kotlin) health-companion app where a virtual pet responds to missed medication doses, concerning vitals, and skipped logging, backed by an offline **Room** database and **WorkManager** background reminders
- Integrated **LLMs via OpenRouter** to power a context-aware chat assistant and auto-build each user's tracked metrics, parsing their conditions and medications into structured **JSON** vital signs

PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

Trust No Ghost - 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped Trust No Ghost for **60+ users** at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Harnessed Unity and C# to develop expanding maze designs, AI pathfinding algorithms, and dynamic moving mechanics